

Khawaja Yunus Ali University

School of Science and Engineering
Department of Computer Science and Engineering

Final Examination: Fall 2020

Program: B.Sc. in CSE (Batch – 11th) 1st Year 1st Semester

Course Title: Physics-I

Course Code: PHY 1101

Time: 10 Minutes

Total Marks: 10 X 1 = 10

Student ID:	Batch:	Session:
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Part 1: MCQ

[These are single best answer type of questions. Each of the following question or statement has five suggested answers or completions. Write down the most appropriate one on your answer script.]

- The maximum displacement of a simple pendulum is called the of the pendulum.
 - frequency
 - time period
 - velocity
 - acceleration
 - amplitude
- What is the unit of the force constant?
 - Nm
 - Nm^{-1}
 - kgm^{-1}
 - Am^{-1}
 - Sec
- In case of SHO, Angular frequency, $\omega = ?$
 - K/m
 - m/K
 - $\sqrt{K/m}$
 - $\sqrt{m/K}$
 - K^2/m^2
- If two bodies are connected by a massless spring to create a two-body system and one of the bodies having mass of $4g$, what is the possible amount of reduced mass of that system?
 - $10g$
 - $8g$
 - $6g$
 - $4g$
 - $2g$
- Which is the correct equation of the maximum kinetic energy of SHO?
 - $KE_{max} = \frac{1}{2}K(A^2 - x^2)$
 - $KE_{max} = \frac{1}{2}K(x^2 - A^2)$
 - $KE_{max} = \frac{1}{2}K(\omega^2 - x^2)$
 - $KE_{max} = \frac{1}{2}KA^2$
 - $KE_{max} = \frac{1}{2}K\omega^2$

P.T.O.

6. RMS value of the alternating current is about of the peak value of it.
- a) 63.7%
 - b) 67.3%
 - c) 70.7%
 - d) 77.0%
 - e) 82.5%
7. What is the frequency of this alternating current: $i = 200 \sin 100\pi t$?
- a) 50Hz
 - b) 100Hz
 - c) 120Hz
 - d) 200Hz
 - e) 220Hz
8. Which of the following devices is used in mobile phone charger?
- a) Ammeter
 - b) Photo diode
 - c) Heat sensor
 - d) Step-down transformer
 - e) Step-up transformer
9. The number of charges per unit volume of a conductor is determined by
- a) Ampere's law
 - b) Biot-Savart Law
 - c) Faraday's law
 - d) lens's law
 - e) Hall effect
10. Current flowing solenoid act as a
- a) magnet.
 - b) capacitor.
 - c) transistor.
 - d) battery.
 - e) DC generator.

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Time: 1 hour and 50 minutes

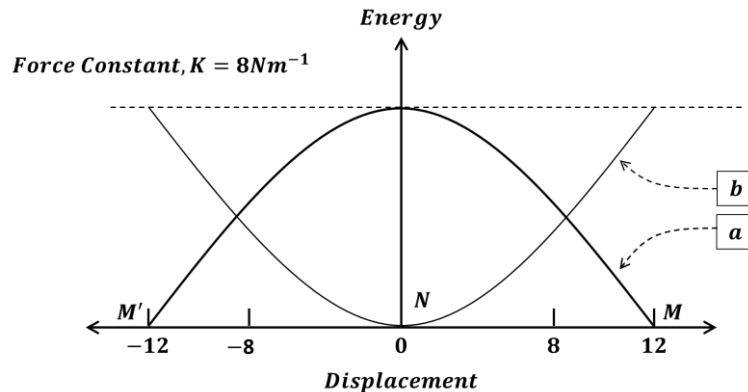
Total Marks: 5×6=30

Part 2: SAQ

[Answer Question No. 1 and any four (04) from the rest]

[The figures in the right margin indicate full marks for the respective question]

1. See the diagram for energy of SHM and answer the questions that follows:



- | | |
|---|---|
| a) Give the name of the curves a and b. | 1 |
| b) Identify the points for the maximum and minimum values of the velocity and acceleration. | 2 |
| c) Prove that the total energy at point M equal to the potential energy. | 3 |
| | |
| 2. a) Investigate the simple harmonic oscillator and indicate the properties of it at different position. | 3 |
| b) If a particle of mass 0.05kg executing simple harmonic motion with frequency 50Hz and if its amplitude and maximum velocity are 0.004m and 0.2 ms ⁻¹ respectively. Calculate its average kinetic energy and the time period of motion. | 3 |
| | |
| 3. a) Define the force Constant. | 1 |
| b) Illustrate the damped oscillation graphically. | 2 |
| c) Two bodies are connected by a massless spring of force constant 1.5dynes/cm. If the reduced mass of the system be 2g and one of the bodies having mass of 10g, then calculate the time period of the oscillation and the mass of other body connected to the system. | 3 |
| | |
| 4. a) Write a short note on “self-induction”. Define 1 henry. | 3 |
| b) Can a transformer be used with DC? – Justify your answer. | 3 |
| | |
| 5. a) State the Ampere’s law. | 1 |
| b) What is hall effect? Mention the applications of it. | 3 |
| c) Magnetic flux of $3 \times 10^{-3}wb$ crosses through a coil having 500 turns. The flux decreases to $1.5 \times 10^{-3}wb$ in 0.015sec. What is the induced e.m.f. ? | 2 |
| | |
| 6. a) Draw the graphical representation of AC and DC. | 1 |
| b) Define form factor of AC. | 2 |
| c) The peak value of an alternating current is 20A and its frequency is 50hz. Find the root mean square value of the current. How long will take to reach the peak from the zero? | 3 |